1 The Relationship Between the Ranks of a Matrix and Its Transpose

To investigate the relationship between the ranks of a matrix and its transpose, we first establish the following result.

Let $A \in M_{m\times n}$

Theorem 1. $\text{rank}(A) = \text{rank}(A^TA)$

Proof. Since A^TA is an $n \times n$ matrix, the rank-nullity theorem implies that

$$\text{rank}(A) = n - nulity(A)$$

$$\text{rank}(A^TA) = n - nulity(A^TA)$$

Suppose that $x \in \ker(A)$. By the definition of the null space, $Ax = 0$. Consequently, $A^TAx = 0$, and hence $x \in \ker(A^TA)$

Therefore, $\ker(A) \subseteq \ker(A^TA)$.

Conversely, suppose that $x \in \ker(A^TA)$. Then, $A^TAx = 0$. Hence,

$$x^TA^TAx = (Ax)^TAx = \|Ax\|^2 = 0$$

Thus, $x \in \ker(A)$. Therefore, $\ker(A^TA) \subseteq \ker(A)$. Consequently, $\ker(A) = \ker(A^TA)$. Equivalently, $nulity(A) = nulity(A^TA)$.

Combining the two inclusions, we conclude that $\text{rank}(A) = \text{rank}(A^TA)$. □

Theorem 2. $\text{rank}(A) = \text{rank}(A^T)$

Proof. Let B be an arbitrary $m \times n$ matrix. Suppose that $x \in \text{Col}(B^TB)$. Then there exists $y \in \mathbb{R}^n$ such that $x = B^TBy$.

$$x = B^TBy = B^T(By)$$

Hence, $x \in \text{Col}(B^T)$. Thus $\text{Col}(B^TB) \subseteq \text{Col}(B^T)$. This implies tha $\text{rank}(B^TB) \leq \text{rank}(B^T)$.

By Theorem 1, we have $\text{rank}(B) = \text{rank}(B^TB)$. Consequently, $\text{rank}(B) \leq \text{rank}(B^T)$ Since

B was arbitrary, we can apply the same argument to B^T. Thus,

$$\text{rank}(B^T) \leq \text{rank}((B^T)^T)$$

Since, $(B^T)^T = B$, it follows that $\text{rank}(B^T) \leq \text{rank}(B)$. Combining the two inequalities, we conclude that

$$\text{rank}(B) = \text{rank}(B^T)$$

Since B was arbitrary, the result holds for A.

□

2 Conditions for the Existence of Left and Right Inverses

To establish the conditions for the existence of left and right inverses, we introduce the following notation:

Let $A \in M_{m\times n}$

Definition 1.

$$\text{Col}(A) = \{Ax \in \mathbb{R}^m : x \in \mathbb{R}^n\} \quad \text{Col}(A) \subseteq \mathbb{R}^m \quad \textit{then} \quad \text{rank}(A) \leq m$$
$$N(A) = \{x \in \mathbb{R}^n : Ax = 0 \in \mathbb{R}^m\} \quad N(A) \subseteq \mathbb{R}^n \quad \textit{then} \quad \textit{nulity}(A) \leq n$$

Theorem 3. $n = \text{rank}(A) + dim(N(A))$

which implies that

$$n = \text{rank}(A) \leq m \iff \textit{nulity}(A) = 0$$

$$\text{rank}(A) = m \iff m \leq n$$

Proposition 1. *A has left inverse* $\implies N(A) = \{0\}$

Proof. Suppose that $A_l^{-1} \in M_{n\times m}$ is a left inverse of A. Then

$$A_l^{-1}A = I_n.$$

Let $x \in N(A)$. Since $Ax = 0$, it follows that

$$\begin{aligned} x &= I_n x \\ &= A_l^{-1} A x \\ &= A_l^{-1}(0) \\ &= 0 \end{aligned}$$

Hence,

$$N(A) = \{0\}.$$

Furthermore, by the rank-nullity theorem,

$$n = \text{rank}(A) \leq m$$

□

Proposition 2. *If $N(A) = \{0\}$, then A has left inverse*

Proof. Suppose that $A^T Ax = \vec{0}$, then,

$$\begin{aligned} x^T A^T A x &= (Ax)^T Ax \\ &= \|Ax\|^2 \\ &= Ax \\ &= \vec{0} \end{aligned}$$

Therefore, $N(A^T A) = N(A) = \{0\}$.

Since $A^T A$ is an $n \times n$ square matrix, the rank-nullity theorem implies that $\text{rank}(A^T A) = n$.

$$(A^T A)^{-1} A^T A = ((A^T A)^{-1} A^T) A = I_n$$

Therefore,

$$(A^T A)^{-1} A^T \text{ is a left inverse of A}$$

□

Proposition 3. *If $\text{rank}(A) = m$, then A has a right inverse.*

Proof. Let y be an arbitrary vector in $\mathbb{R}^m$.

Since $\text{rank}(A) = m$, there exists $x \in \mathbb{R}^m$ such that $Ax = y$, which means $\text{Col}(A) = \mathbb{R}^m$.

Since $\text{rank}(A) = \text{rank}(A^T) = m$ and $m = \text{rank}(A^T) + nulityA^T$.

$$N(A^T) = \{0\}$$

Suppose that $AA^Tx = 0$. Then

$$x^TAA^Tx = \|A^Tx\|^2 = 0$$

$$N(AA^T) \subseteq N(A^T)$$

Since $N(A^T) = \{0\}$, it follows that $N(AA^T) = \{0\}$.

By using the rank-nulity theorem,

$$\text{rank}(AA^T) = m$$

Because AA^T is a square matrix, it is therefore invertible.

$$AA^T(AA^T)^{-1} = A(A^T(AA^T)^{-1}) = I_m$$

Hence, $A^T(AA^T)^{-1}$ is right inverse of A □